



# *Smart Door Lock System with Firebase*

Group 5

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# *Project Overview*

IoT-based smart door lock system  
Uses Raspberry Pi as main controller  
Multi-layer authentication for security





# *Problem Statement*

Conventional locks are vulnerable  
No access monitoring  
Risk of key duplication





# *System Objectives*

Enhance security using multi-factor authentication

This objective is achieved by combining face recognition and RFID authentication, ensuring access is granted only when both identity verification methods are successfully validated.

Automate door access

Door access is automated using sensors and a relay-controlled solenoid lock, allowing the system to unlock and lock the door automatically without manual keys.

Provide access logging and monitoring

The system records every access attempt to Firebase Realtime Database, enabling real-time monitoring and review of access history.





# *System Architecture*

## Raspberry Pi

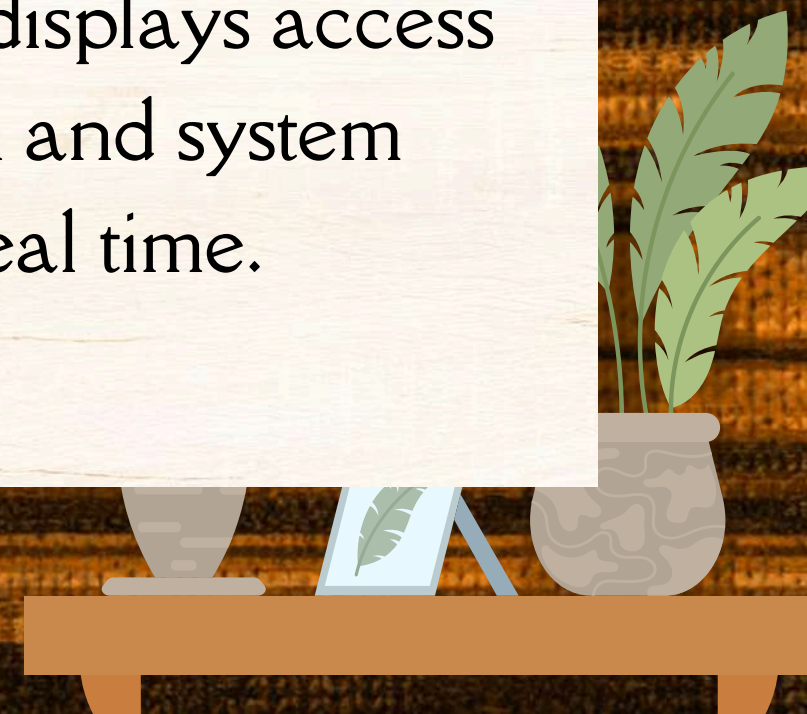
Raspberry Pi acts as the main controller of the system, processing input data from sensors, handling authentication logic, and controlling the door lock mechanism.

## Sensors

The system uses an ultrasonic sensor (HC-SR04) to detect user presence, a camera module to capture facial images for face recognition, and an RFID reader (MFRC522) to read card data for identity verification.

## Firebase and HTML

Firebase is used for data storage and access logging, while the HTML-based dashboard displays access information and system activity in real time.



# *Implementation*





# *Data Analysis*

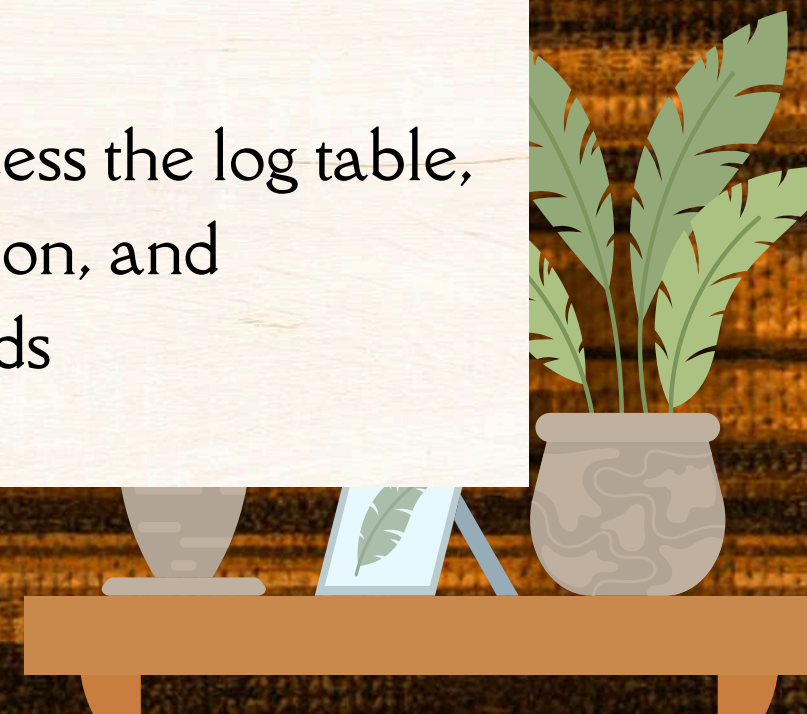
## Authentication Flow

Ultrasonic detects presence  
Face recognition verification  
RFID card authentication  
Access granted or denied

## Firebase Integration

Logs access events  
Stores UID, status, time  
Supports real-time monitoring

Dashboard Features: Access the log table,  
providing chart visualization, and  
refreshing every 15 seconds



Four decorative leaves hang from the top left corner. Two are light green and two are light yellow, all with a braided stem and a small white bead at the top.

## *Advantage & Disadvantage*

### Advantage:

- Enhanced security
- Automation & convenience
- Real-time monitoring
- Efficient operation

### Disadvantage:

- Dependent on power supply
- Sensitive to lighting conditions
- Limited user scalability
- No anti-spoofing detection



Four decorative hanging leaves, two green and two yellow, are suspended from the top left corner of the slide.

# *Conclusion*

The Smart Door Lock System with Firebase successfully integrates IoT technology and multi-layer authentication to enhance security and automate door access. By combining ultrasonic detection, face recognition, and RFID authentication, the system improves both safety and user convenience. Despite limitations such as power dependency and environmental sensitivity, this system provides a strong foundation for future development of smarter and more secure access control solutions.

A decorative potted plant with green leaves is situated in the bottom right corner of the slide.



# *Thank You*

Presentation by Fauget Group